```
C:\Users\owner\Documents\GitHub\MageZero\.venv\Scripts\python.exe C:\Users\owner\Documents\GitHub\MageZero\test.py

ignore list size: 1979605
[opponent filter] scanned 244871 samples  kept 202872 (removed 41999 opponent states).
Loaded checkpoint from models/Standard-MonoU/ver1/model.pt
Validation loss:  priority_A_loss=0.013  priority_B_loss=0.000 choose_target_loss=0.144 choose_use_loss=0.039 value_loss=0.046 avg_total_loss=0.243 decision_states=111267
Test priority_A_accuracy=0.848
--- Policy Confusion Matrix (True \ Predicted) ---
True |   0   1   2   3   4   5   6   7   8   9  10  11  12  13  14  15  16  17  18  19  20  21  22  23  24  25  26  27  28  29  30  31
-------------------------------------------------------------------------------------------------------------------------------------
   0 |32679  38  68 534 234 130 591 140 381 135 1891199   0  84  19  74  54   0  21  12  37 195 225   0   0   0   0   0   0   0   0   0
   1 |  26 500  11   0   0   0   1   0  60   1   1  43   0   0   0   0   0   0   1   2   3   0  11   0   0   0   0   0   0   0   0   0
   2 |  70  14 407   0   0   0   0   0  46   0   1  24   0   0   0   0   0   0   0   2   4   0  14   0   0   0   0   0   0   0   0   0
   3 |1777   0   31343   0   1  34   0   2   3   7   2   0   0   0   0   0   0   2   0   0   0   4   0   0   0   0   0   0   0   0   0
   4 | 184   0   0   01399  29   0   0   0   0   0   0   0   0   0  10  10   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0
   5 |  30   0   0   1  321085   0   0   0   0   0   0   0   0   0   1   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0   0
   6 | 779   0   4   3   3   22832   0   3   0   4   7   0   0   0   1   1   0   1   0   0   1   1   0   0   0   0   0   0   0   0   0
   7 | 418   2   5   3   1   0   3 446   5   0   2   4   0   9   0   0   0   0   1   2   3   6  14   0   0   0   0   0   0   0   0   0
   8 | 179  36   2   0   0   0   4   02983   4   0 109   0   0   0   0   0   1   3  14  13   1  21   0   0   0   0   0   0   0   0   0
```

```
19    9 |  81   1   1   0   1   0   3   2   5 456   0   1   0
      0   0   0   1   0   0   0   2   1   2   0   0   0   0
   0   0   0   0   0
20   10 | 435   1   0   5   0   0   5   0  10   1 949   1   0
      0   0   1   0   0   1   0   0   7   9   0   0   0   0
   0   0   0   0   0
21   11 | 124  31  14   0   0   0   1   0 114   0   015174   0
      0   0   0   0   7  19  19  16   0  61   0   0   0   0
   0   0   0   0   0
22   12 |   0   0   0   0   0   0   0   0   0   0   0   0   0
      0   0   0   0   0   0   0   0   0   0   0   0   0   0
   0   0   0   0   0
23   13 | 245   4   0   1   1   0   1   0   0   1   2   0   0
   143   0   0   0   0   0   0   0   0   4   0   0   0   0   0
      0   0   0   0
24   14 |  80   0   0   0   0   0   0   0   0   0   0   0   0
      0  48   0   0   0   0   0   0   0   0   0   0   0   0
   0   0   0   0   0
25   15 |  77   0   0   0   5   1   0   0   0   0   0   0   0
      0   0 520   7   0   0   0   0   0   0   0   0   0   0
   0   0   0   0   0
26   16 | 144   0   0   0  10   0   0   0   0   0   0   0   0
      0   0  34 498   0   0   0   0   0   0   0   0   0   0
   0   0   0   0   0
27   17 |   0   0   0   0   0   0   0   0   0   0   0   2   0
      0   0   0   0   7   1   0   0   0   0   0   0   0   0
   0   0   0   0   0
28   18 |   1   0   0   0   0   0   0   0   2   0   0   2   0
      0   0   0   0   0 477  22   0   0   1   0   0   0   0
   0   0   0   0   0
29   19 |   2   2   0   0   0   0   0   0   8   0   0  15   0
      0   0   0   0   0  82 315   0   0   0   0   0   0   0
   0   0   0   0   0
30   20 |  13   6   1   0   0   0   1   0   4   0   0  23   0
      0   0   0   0   0   0   0 416   1   4   0   0   0   0
   0   0   0   0   0
31   21 | 744   8  14   4   3   0   5   4  30   0   8   3   0
      0   0   0   0   0   0   0   2 555  11   0   0   0   0
   0   0   0   0   0
32   22 | 181  12  12   0   0   0   6   1 102   4   3 129   0
      4   0   0   0   0   2  10   8   61706   0   0   0   0
   0   0   0   0   0
33   23 |   0   0   0   0   0   0   0   0   0   0   0   0   0
      0   0   0   0   0   0   0   0   0   0   0   0   0   0
   0   0   0   0   0
34   24 |   0   0   0   0   0   0   0   0   0   0   0   0   0
```

```
34    0   0   0   0   0   0   0   0   0   0   0   0   0   0
   0   0   0   0   0
35   25 |   0   0   0   0   0   0   0   0   0   0   0   0   0
      0   0   0   0   0   0   0   0   0   0   0   0   0   0
   0   0   0   0   0
36   26 |   0   0   0   0   0   0   0   0   0   0   0   0   0
      0   0   0   0   0   0   0   0   0   0   0   0   0   0
   0   0   0   0   0
37   27 |   0   0   0   0   0   0   0   0   0   0   0   0   0
      0   0   0   0   0   0   0   0   0   0   0   0   0   0
   0   0   0   0   0
38   28 |   0   0   0   0   0   0   0   0   0   0   0   0   0
      0   0   0   0   0   0   0   0   0   0   0   0   0   0
   0   0   0   0   0
39   29 |   0   0   0   0   0   0   0   0   0   0   0   0   0
      0   0   0   0   0   0   0   0   0   0   0   0   0   0
   0   0   0   0   0
40   30 |   0   0   0   0   0   0   0   0   0   0   0   0   0
      0   0   0   0   0   0   0   0   0   0   0   0   0   0
   0   0   0   0   0
41   31 |   0   0   0   0   0   0   0   0   0   0   0   0   0
      0   0   0   0   0   0   0   0   0   0   0   0   0   0
   0   0   0   0   0
42 ------------------------------------------------------------
   -
43 No priority B samples in test set to calculate accuracy.
44 Test choose_target_accuracy=0.548
45 --- Policy Confusion Matrix (True \ Predicted) ---
46 True |   0   1   2   3   4   5   6   7   8   9  10  11  12
     13  14  15  16  17  18  19  20  21  22  23  24  25  26
   27  28  29  30  31
47 ------------------------------------------------------------
   ------------------------------------------------------------
   ----------------
48    0 |3639   0   0   5  25  26  24  10   5   9 116  22  10
   162   6  13   8  12   5   9  32 235  56  90  55  62  24   3
     20  69  29  69
49    1 |   0   0   0   0   0   0   0   0   0   0   0   0   0
      0   0   0   0   0   0   0   0   0   0   0   0   0   0
   0   0   0   0   0
50    2 |   0   0   0   0   0   0   0   0   0   0   0   0   0
      0   0   0   0   0   0   0   0   0   0   0   0   0   0
   0   0   0   0   0
51    3 |  23   0   0  14   1   3   6   1   4   0   8   2   0
     91   0   2   1   0   0   2   2   0   0   0   0   0   0
   0   0   0   0   0
```

```
52    4 |  46   0   0   0  59   0  11   1   5   4  15   5   3
     93   1   6   4   3   3   6   8   2   0   0   1   0   0
   0   0   0   0   0
53    5 | 104   0   0   4   2 122  49   5   6  16  12   5   7
   428  11  13   9   2   0   8   9   0   0   1   0   0   0   0
      0   0   0   0
54    6 | 177   0   0   5   0  12 591   3   4  16   8  15   9
   434   9  19  20   1   0   4   9   0   1   0   0   0   0   0
      0   0   0   0
55    7 |  80   0   0   1   0   7  24  31   5   7  20   3   2
   174   1  12   7   4   0   3   8   0   0   0   0   0   0   0
      0   0   0   0
56    8 |  85   0   0   6   1  14  29   5  72  10  32   4  10
   300   5   3   4   3   0   6   5   0   0   0   0   0   0   0
      0   0   0   0
57    9 |  69   0   0   1   0   8  40   2   6 121   6  12   6
   157  12  13  10   3   0   7   2   0   0   0   0   0   0   0
      0   0   0   0
58   10 | 102   0   0   4   6   7  17   7   7   7 415  13   2
   259   4   8   3   3  10   3  18   5   3   0   7   2   0   0
      0   3   0   0
59   11 | 107   0   0   1   5   7  20   1   1  10  10 185   3
   167   5  14  15   0  13   4  25   1   1   0   5   1   0   0
      0   2   0   2
60   12 |  81   0   0   2   0   5  16   0   1   3   3   3  49
     87   2   6   4   2   0   5   1   0   0   0   0   0   0
   0   0   0   0   0
61   13 | 692   0   0   5   6  32  91  11  17  21  50  21
   203244  22  49  28   7   0  16  24   0   0   0   0   0   0
      0   0   1   0   0
62   14 |  49   0   0   0   0   0  21   3   2   4   6  12   0
     57  88   8   4   1   0   4   2   0   0   0   0   0   0
   0   0   0   0   0
63   15 | 100   0   0   2   0   9  38   3   1   5   5   6   7
   106  12 153  16   2   0   1   3   0   0   0   0   0   0   0
      0   0   0   0
64   16 |  85   0   0   0   0   4  47   0   4   9   4   9   5
   130   9  24 196   0   0   2   1   0   0   0   0   0   0   0
      0   0   0   0
65   17 |  47   0   0   0   0   0   5   0   0   0   8   4   2
     37   0   0   1  23  10   0   5   1   0   0   0   0   0
   0   0   0   1   0
66   18 |   2   0   0   0   0   0   0   0   0   0   1   2   0
      0   0   0   0   2  27   0   3   0   0   0   0   0   0
   0   0   0   0   0
67   19 |  71   0   0   1   3   5  20   1   1   8  13   3   4
```

```
 67 102   4   4   5   1   0  54   3   1   0   0   0   0   0
    0   0   0   0   0
 68   20 | 111   0   0   0   2   5  15   2   2   2  52  11   8
     164   1   7   4   3   4   1 173   4   2   0   0   1   0
       0   0   0   0   2
 69   21 |  13   0   0   0   0   0   0   0   0   0   0   0   0
       0   0   0   0   1   0   0   1 366   9   1   3   5   4
       1   4   1   7  11
 70   22 | 111   0   0   0   0   0   0   0   0   0   0   0   0
       0   0   0   0   0   1   0   1  19  81  15  18   7   5
       0   2   0   1   5
 71   23 |  53   0   0   0   0   0   0   0   0   0   2   1   0
       0   0   0   0   0   1   0   1  15   5 211  16  17  15
       0   2   1   1  12
 72   24 |   7   0   0   0   0   0   0   0   0   0   1   2   0
       0   0   0   0   2   4   0   3  14   4   4 249  24   9
       1   3   4   6   6
 73   25 |   3   0   0   0   0   0   0   0   0   0   1   0   0
       0   0   0   0   0   0   0   3  65   1   3  15 243  12
       0   5   0   4  11
 74   26 |  40   0   0   0   0   0   0   0   0   0   4   7   0
       0   0   0   0   2   1   0   1   3  13   7  21  26 152
       0   1   0   5  19
 75   27 |   5   0   0   0   0   0   0   0   0   0   1   0   0
       0   0   0   0   2   0   0   0   2   2   0   1   0   0
       8   0   5   1   0
 76   28 |   4   0   0   0   0   0   0   0   0   0   1   0   0
       0   0   0   0   0   0   0   0   4   0   0   2   1   2
       0  46   0   0   0
 77   29 |  33   0   0   0   0   0   0   0   0   0   0   0   0
       0   0   0   0   0   1   0   0   2   0   2   2   3   1
       0   0  76   2   3
 78   30 |  31   0   0   0   0   0   0   0   0   0   1   6   0
       0   0   0   0   1   0   0   0   6   0   2  10  11   1
       3   0  10 100   8
 79   31 |   3   0   0   0   0   0   0   0   0   0   0   3   0
       0   0   0   0   0   0   0   0   5   2   3   1   2   1
       0   0   1   2 169
 80 ------------------------------------------------------------
    --
 81 Test choose_use_accuracy=0.882
 82 --- Policy Confusion Matrix (True \ Predicted) ---
 83   True   |       0       1
 84 --------------------------
 85        0 |    3698     998
 86        1 |     740    9270
```

```
---------------------------------------------------------------
--

Process finished with exit code 0

```